

Supplementary Material

A transdiagnostic dimensional map as a self-calibrating measurement instrument: projection, information, and value of information under structured missingness

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This document gives the full derivations that the Methods section of the main text (§2.1–§2.7) states as results and defers here. It is self-contained: every object it uses is defined below, and cross-references of the form “Eq. (7)” point to the numbered equations of the main manuscript. Sections S1 and S2 develop the measurement model and the projection machinery from first principles; Sections S3–S7 derive in full the five instrument properties that are the paper’s contribution — the loading as a measurement slope, the information accounting and per-family Fisher information, adaptive sampling and value of information, the archetypal summary and its reconstruction fidelity, and the calibration protocol. Propositions are proved; everything reported in the main text is an evaluation of the objects defined here.

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S1 The measurement model from first principles

S1.1 Generative factor model and local independence

Index patients $i = 1, \dots, N$, indicators $j = 1, \dots, J$, specific factors $k = 1, \dots, K$ and one general factor G . Each patient carries a factor vector $f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \in \mathbb{R}^{K+1}$ with

$$f_i \sim \mathcal{N}_{K+1}(\mathbf{0}, \Phi), \quad \text{diag}(\Phi) = \mathbf{1}, \quad (\text{S1})$$

the factor *correlation* matrix Φ carrying a unit diagonal to fix the latent scale. Each indicator is a linear reflection of the factors plus item-specific noise through a single linear predictor,

$$\eta_{ij} = \alpha_j + \lambda_{jG} G_i + \sum_{k=1}^K \lambda_{jk} D_{ik} + \beta_j^\top c_i, \quad (\text{S2})$$

with item intercept α_j , general loading λ_{jG} , specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\lambda_G \mid \Lambda_D] \in \mathbb{R}^{J \times (K+1)}$), and item-level covariates c_i that adjust an indicator’s mean but are never factor indicators. This is Eq. (1) of the main text. Stacking one patient’s items,

$$x_i = \mu_i + \lambda_G G_i + \Lambda_D D_i + \varepsilon_i, \quad \mu_i = \alpha + B c_i, \quad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi), \quad (\text{S3})$$

with diagonal residual covariance $\Psi = \text{diag}(\sigma_1^2, \dots, \sigma_J^2)$.

Local independence. Given a patient’s latent scores the items are conditionally independent,

$$p(x_i \mid f_i, \Theta) = \prod_{j \in O_i} p(x_{ij} \mid f_i, \Theta), \quad (\text{S4})$$

i.e. the factors are the only reason items covary. In the Gaussian linear model this is identical to “ Ψ is diagonal”: local independence and diagonal residual covariance are two faces of one assumption. It is also why filling missing cells is forbidden — an invented value injects covariance the model would mistake for a factor. Because f_i is unobserved we integrate it out and multiply across independent patients,

$$L(\Theta) = \prod_{i=1}^N \int \left[\prod_{j \in O_i} p(x_{ij} \mid f_i, \Theta) \right] \mathcal{N}(f_i; \mathbf{0}, \Phi) \mathrm{d}f_i. \quad (\text{S5})$$

Equation (S2) is a bifactor exploratory structural-equation model: every indicator may load on G and on its home specific factor, while off-home cells are governed by a sparsity-inducing prior (§S1.4) that keeps the map mostly simple-structure yet lets a few evidence-backed cross-loadings emerge. The fitted map has $K+1 = 8$ factors: G (overall burden) and seven specific axes — cognition, immunometabolic, sleep, mania/activation, suicidality, developmental-risk and substance. The fitted map carries 109 indicators (88 in the Gaussian-copula continuous block, 21 in the explicit non-Gaussian block).

S1.2 Mixed-likelihood observation model and the Gaussian copula

Coercing variables onto a common scale would manufacture covariance and bias the very factors being estimated. Each indicator instead keeps the observation density appropriate to its type, all sharing the linear predictor (S2):

$$\text{continuous / } z\text{-score: } X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (\text{S6})$$

$$\text{skewed laboratory: } \log X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (\text{S7})$$

$$\text{ordinal } (C_j \text{ levels}): \Pr(X_{ij} \leq c) = \text{logit}^{-1}(\tau_{jc} - \eta_{ij}), \quad \tau_{j1} < \dots < \tau_{j,C_j-1}, \quad (\text{S8})$$

$$\text{binary: } X_{ij} \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_{ij})), \quad (\text{S9})$$

$$\text{count: } X_{ij} \sim \text{NegBinom}(\mu_{ij} = e^{\eta_{ij}}, r_j). \quad (\text{S10})$$

The reported map replaces the raw Gaussian likelihood on the continuous block with a semiparametric Gaussian-copula model: each continuous indicator is mapped to a latent Gaussian score by its frozen empirical distribution function $\hat{F}_j$, $z_{ij} = \Phi_{\mathcal{N}}^{-1}(\hat{F}_j(X_{ij}))$, and the factor model is fit on z_{ij} . Because the rank transform is monotone it alters none of the identification arguments below. Writing $F_j(\cdot)$ for indicator j 's density and $O_i \subseteq \{1, \dots, J\}$ for its observed indicators, the observed-data log-likelihood sums over observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^N \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j), \quad (\text{S11})$$

which is Eq. (3) of the main text: a missing cell contributes no term and no completed $N \times J$ matrix is ever formed.

S1.3 Missing data is the marginal property, not imputation

Proposition 1 (Marginalization). *For patient i with observed continuous set $C_i \subseteq O_i$, the model-implied distribution of the observed sub-vector is obtained by deleting the unobserved coordinates from (μ, Σ) :*

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i}, \Sigma_{C_i}), \quad \Sigma_{C_i} = \Lambda_{C_i} \Phi \Lambda_{C_i}^\top + \Psi_{C_i}, \quad (\text{S12})$$

with Λ_{C_i} the observed-row loadings and $\Psi_{C_i} = \text{diag}(\sigma_j^2)_{j \in C_i}$.

Proof. A foundational property of the multivariate normal is that marginals are read off by deleting coordinates: if $x \sim \mathcal{N}(\mu, \Sigma)$ then any sub-vector x_C is $\mathcal{N}(\mu_C, \Sigma_{CC})$. Apply this to $x_i \sim \mathcal{N}(\mu_i, \Sigma)$ with $\Sigma = \Lambda \Phi \Lambda^\top + \Psi$ (Proposition 3); the submatrix Σ_{CC} retains exactly the observed rows and columns of Λ and the observed diagonal of Ψ , giving (S12). $\square$

Hence “integrating out” the missing cells is not an extra step but the marginal property itself; summing $\log \mathcal{N}(\cdot)$ over each patient’s observed sub-vector is precisely the full-information maximum-likelihood (FIML) objective for incomplete multivariate-normal data. This is the formal justification of the no-imputation invariant (S11): any completed matrix would inject between-cell covariance that (S4) would misattribute to a latent factor.

S1.4 Identifiability: rotation invariance and how the prior resolves it

Proposition 2 (Rotation invariance of the likelihood). *For any invertible $M \in \mathbb{R}^{(K+1) \times (K+1)}$, the reparameterization $\Lambda^* = \Lambda M$, $\Phi^* = M^{-1} \Phi M^{-\top}$, $f^* = M^{-1} f$ leaves the implied covariance, and hence the Gaussian likelihood, unchanged:*

$$\Lambda^* \Phi^* \Lambda^{*\top} + \Psi = \Lambda M (M^{-1} \Phi M^{-\top}) M^\top \Lambda^\top + \Psi = \Lambda \Phi \Lambda^\top + \Psi = \Sigma. \quad (\text{S13})$$

Proof. Immediate from $MM^{-1} = I$: the inner $M(M^{-1}\Phi M^{-\top})M^{\top}$ telescopes to Φ . Since x_i is determined in distribution by (μ_i, Σ) and Σ is invariant, so is the likelihood. $\square$

A factor model is therefore not identified from the likelihood alone. Confirmatory factor analysis breaks this by fixing loadings to exactly zero; the present model breaks it with the prior. Each (indicator, factor) cell is assigned a role and prior

$$\lambda_{jk} \mid \rho(j, k) \sim \begin{cases} \mathcal{N}_+(0.70, 0.25^2) & \text{PRIMARY (HOME CELL),} \\ \text{regularized horseshoe (S15)} & \text{PLAUSIBLE CROSS,} \\ \mathcal{N}(0, 0.05^2) & \text{UNLIKELY,} \\ \delta_0 & \text{FORBIDDEN,} \end{cases} \quad (\text{S14})$$

with $\mathcal{N}_+$ truncated to $(0, \infty)$ and δ_0 fixing the loading at zero. The likelihood is rotation-invariant but the prior is not: a rotation moves a loading the prior wanted near zero away from zero, lowering $p(\Lambda)$, so the posterior $p(\Lambda \mid X) \propto p(X \mid \Lambda) p(\Lambda)$ concentrates at the orientation where small loadings align with the soft-zero priors and home loadings are positive and substantial; the positive truncation removes the residual sign-flip. The plausible-cross cells use a regularized (“Finnish”) horseshoe,

$$\lambda_j = z_j \tau \tilde{\eta}_j, \quad \tilde{\eta}_j = \sqrt{\frac{c^2 \eta_j^2}{c^2 + \tau^2 \eta_j^2}}, \quad z_j \sim \mathcal{N}(0, 1), \quad \eta_j \sim \text{HalfCauchy}(1), \quad \tau \sim \text{HalfNormal}(\tau_0), \quad (\text{S15})$$

with $\tau_0 = 0.05$ and slab $c = 0.30$: default-off (global τ shrinks the set toward zero), evidence-on (heavy-tailed local η_j lets a genuine cross-loading escape), and magnitude-capped ($\tilde{\eta}_j \rightarrow c/\tau$, so an escaped loading stays small). This is the honest treatment of weak signal in a factor with few indicators.

Corollary 1 (Flat-prior confirmation). *Under a flat loading prior (identification constraints only) the posterior mode equals the maximum-likelihood (FIML) solution. A prior-free refit landing on the same loadings therefore demonstrates that the structure is recovered from the data, not manufactured by the priors; empirically this refit reproduced the map to Tucker congruence $\varphi = 1.00$.*

S1.5 The covariance identity and the correlated- G estimand

Proposition 3 (Bifactor covariance). *Under (S3) with $\text{Var}(G_i) = 1$, $\text{Cov}(D_i) = \Phi_D$, $\mathbb{E}[\varepsilon_i \varepsilon_i^{\top}] = \Psi$ diagonal, and the bifactor orthogonality $\mathbb{E}[G_i D_i^{\top}] = \mathbf{0}$ together with $(G_i, D_i) \perp \varepsilon_i$,*

$$\Sigma = \text{Cov}(x_i) = \underbrace{\lambda_G \lambda_G^{\top}}_{\text{general: rank one}} + \underbrace{\Lambda_D \Phi_D \Lambda_D^{\top}}_{\text{specific factors}} + \underbrace{\Psi}_{\text{private noise}} = \Lambda \Phi \Lambda^{\top} + \Psi. \quad (\text{S16})$$

Proof. Expand $\Sigma = \mathbb{E}[(x_i - \mu_i)(x_i - \mu_i)^{\top}]$ with $x_i - \mu_i = \lambda_G G_i + \Lambda_D D_i + \varepsilon_i$. The cross-terms $\lambda_G \mathbb{E}[G_i D_i^{\top}] \Lambda_D^{\top}$ and $\mathbb{E}[(\lambda_G G_i + \Lambda_D D_i) \varepsilon_i^{\top}]$ vanish by $\mathbb{E}[G_i D_i^{\top}] = \mathbf{0}$ and by independence of factors and noise respectively. The survivors are $\lambda_G \lambda_G^{\top} \text{Var}(G_i)$, $\Lambda_D \Phi_D \Lambda_D^{\top}$ and Ψ , giving (S16). $\square$

The general factor leaves a rank-one footprint: every pair of items covaries through G by exactly $\lambda_{jG} \lambda_{j'G}$. The correlated- G arm (Eq. (4) of the main text) unzips the zero off-diagonal block, freeing $\phi_{Gd} = \text{Corr}(G, D)$, whose immunometabolic entry near zero against far larger cognition and sleep entries is the estimand the companion paper interprets. Here it is used only as a worked example of the instrument.

S2 Projection: the Woodbury reduction and EAP scoring

S2.1 Woodbury reduction of the observed-cell covariance

Evaluating (S12) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity and matrix-determinant lemma reduce it to the $(K+1)$ -dimensional factor space.

Lemma 1 (Woodbury reduction). *With Ψ_{C_i} diagonal and Λ_{C_i} the observed-row loadings,*

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1} \Lambda_{C_i} (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1} \Lambda_{C_i}^\top \Psi_{C_i}^{-1}, \quad (\text{S17})$$

$$\log |\Sigma_{C_i}| = \log |\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}| + \log |\Phi| + \sum_{j \in C_i} \log \sigma_j^2. \quad (\text{S18})$$

Proof. Equation (S17) is the Woodbury matrix identity applied to $\Sigma_{C_i} = \Psi_{C_i} + \Lambda_{C_i} \Phi \Lambda_{C_i}^\top$ with the low-rank term $\Lambda_{C_i} \Phi \Lambda_{C_i}^\top$; (S18) is the matrix-determinant lemma $\log |\Psi + U \Phi U^\top| = \log |\Phi^{-1} + U^\top \Psi^{-1} U| + \log |\Phi| + \log |\Psi|$ with $U = \Lambda_{C_i}$ and Ψ diagonal. $\square$

The inner solve is $\mathcal{O}((K+1)^3)$, independent of $|C_i|$; patients sharing a missingness pattern share the inner $(K+1) \times (K+1)$ factorization, computed once per pattern. This makes full-sample ($N = 9,013$) projection tractable and is Eq. (6) of the main text; the non-Gaussian block retains explicit per-patient latents.

S2.2 EAP factor scores and Bayes/FIML equivalence

Proposition 4 (EAP scores). *For an all-continuous observed pattern, the posterior of f_i given the observed cells is Gaussian with mean and covariance*

$$\hat{f}_i = \mathbb{E}[f_i | X_{i,C_i}] = \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} (X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi, \quad (\text{S19})$$

which is Eq. (7) of the main text.

Proof. Under (S1) and (S3) the joint of (f_i, X_{i,C_i}) is Gaussian with $\text{Cov}(f_i) = \Phi$, $\text{Cov}(X_{i,C_i}) = \Sigma_{C_i}$ and cross-covariance $\text{Cov}(f_i, X_{i,C_i}) = \Phi \Lambda_{C_i}^\top$. The Gaussian conditioning formula $\mathbb{E}[f | X] = \text{Cov}(f, X) \text{Cov}(X)^{-1} (X - \mu_X)$ and $\text{Cov}(f | X) = \text{Cov}(f) - \text{Cov}(f, X) \text{Cov}(X)^{-1} \text{Cov}(X, f)$ give (S19) directly. $\square$

Proposition 5 (Equivalent precision form). *The posterior covariance (S19) admits the precision form*

$$S_i^{-1} = \Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}, \quad (\text{S20})$$

so the posterior precision is the prior precision plus a data term that is a sum of per-item contributions.

Proof. Substitute the Woodbury form (S17) into $S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi$ and simplify; the low-rank correction telescopes to give $S_i = (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1}$. Equivalently, (S20) is the standard Gaussian information form of the posterior of a linear model with prior precision Φ^{-1} and per-observation precision $\Psi_{C_i}^{-1}$ propagated through Λ_{C_i} . Inverting gives the covariance form (S19). $\square$

Equation (S20) is the pivot of the whole instrument: it exhibits the posterior precision as *additive over observed items*, which §S4 turns into the information accounting. When non-Gaussian cells are present the posterior is no longer exactly Gaussian; $\hat{f}_i$ and S_i are then the posterior mean and covariance obtained by MCMC or, for the calibration study, by the Fisher-scoring Laplace approximation of §S7, which reduces to (S19) exactly on all-Gaussian patterns.

Proposition 6 (Bayes/FIML equivalence). *The marginalized Bayesian observed-data model and FIML optimize the same observed-data likelihood (S11) and differ only by the priors (S14). In particular (Corollary 1) the flat-prior MAP equals the FIML estimate, so a standalone FIML estimator adds no independent evidence and confirmation is done in-engine.*

Proof. By Proposition 1 the per-patient marginal density is the incomplete-data multivariate-normal density on C_i , whose log summed over patients is the FIML objective. The posterior is this likelihood times the prior; with a flat prior the log-posterior equals the log-likelihood up to a constant, so its mode is the FIML maximizer. $\square$

Finally, the structure of (S19) makes the instrument’s honesty on unmeasured axes mechanical: on an axis with no observed indicator the corresponding rows of Λ_{C_i} are empty, the update term in (S20) vanishes on that coordinate, and the posterior returns the prior $\mathcal{N}(0, \Phi)$ — a wide interval, never a false-precise point. The per-patient posterior covariance S_i is exported and propagated into every downstream analysis; a patient with one observed indicator on a factor is not treated as equally characterized as one with six.

S3 The loading as a measurement slope

The main text (§2.2) separates two objects that (S2) places on the same line: the per-patient latent *position* f_i , and the fixed *slope* at which a measurement changes as that position moves. This section derives the slope on both the latent and the response scale.

Proposition 7 (Latent-scale slope). *On the linear-predictor scale the sensitivity of indicator j to factor k is exact, constant across patients, and equal to the loading:*

$$\frac{\partial \eta_{ij}}{\partial f_{ik}} = \lambda_{jk}. \quad (\text{S21})$$

Hence the loading matrix Λ is the Jacobian $\partial \eta_i / \partial f_i$ of the measurement with respect to the latent coordinate.

Proof. Differentiate (S2) with respect to f_{ik} ; the intercept, the covariate term $\beta_j^\top c_i$ and the other factor terms do not depend on f_{ik} , leaving λ_{jk} . $\square$

Proposition 8 (Response-scale slope). *For a single-index likelihood with conditional mean $\mathbb{E}[X_{ij} | f_i] = h_j(\eta_{ij})$ (link mean function h_j), the marginal effect of a factor on the indicator and its population average are*

$$\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} = h'_j(\eta_{ij}) \lambda_{jk}, \quad s_{jk} \equiv \mathbb{E} \left[\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} \right] = \kappa_j \lambda_{jk}, \quad \kappa_j = \mathbb{E} [h'_j(\eta_{ij})] \geq 0, \quad (\text{S22})$$

which is Eq. (5) of the main text.

Proof. The chain rule applied to $\mathbb{E}[X_{ij} | f_i] = h_j(\eta_{ij})$ with $\partial \eta_{ij} / \partial f_{ik} = \lambda_{jk}$ (Proposition 7) gives the first identity. Taking the expectation over the population distribution of η_{ij} and factoring out the patient-independent λ_{jk} defines $\kappa_j = \mathbb{E}[h'_j(\eta_{ij})]$, which is non-negative because each family’s mean function h_j is non-decreasing ($h'_j = \sigma^{-2}$ -scaled identity for the Gaussian, $p(1-p)$ for the logit-Bernoulli, e^η for the log-count). $\square$

The loading is therefore *how strongly an item reflects a factor*. It is not how much that item localizes a patient — that is the Fisher information of §S4, and the two coincide only for the Gaussian family, where h'_j is constant. The response-scale average κ_j is what makes a unit-loading rare binary flag a weak *measurement* despite a large λ : $\kappa_j = \mathbb{E}[p_{ij}(1-p_{ij})]$ collapses as the endorsement probability leaves $\frac{1}{2}$.

S4 Information accounting: what localizes a patient

S4.1 Additivity of posterior precision

Proposition 9 (Additive precision). *Under local independence (S4) the posterior precision of the coordinate is the prior precision plus a sum of per-item Fisher-information contributions,*

$$S_i^{-1} = \Phi^{-1} + \sum_{j \in O_i} \mathcal{I}_j(f_i), \quad \mathcal{I}_j(f_i) = -\mathbb{E}_{X_{ij}|f_i} \left[\nabla_{f_i}^2 \log F_j(X_{ij} | \eta_{ij}) \right], \quad (\text{S23})$$

which is Eq. (8) of the main text. Each observed cell adds a positive semidefinite term; nothing an item measures is ever subtracted.

Proof. The log-posterior of f_i given the observed cells is, up to a constant, $\log p(f_i | X_{i,O_i}) = \log \mathcal{N}(f_i; \mathbf{0}, \Phi) + \sum_{j \in O_i} \log F_j(X_{ij} | \eta_{ij})$, the sum over items following from local independence (S4). The negative Hessian in f_i of the prior term is Φ^{-1} ; the negative expected Hessian of each item term is by definition the item's Fisher information $\mathcal{I}_j(f_i)$ about the factors. Summing gives (S23); each $\mathcal{I}_j$ is positive semidefinite as an expected outer product of score vectors (below), so observing an item can only increase precision. For the Gaussian family the expectation is exact and (S23) coincides with (S20); for the other families it is the expected-information (Fisher-scoring) form used in §S7. $\square$

S4.2 Per-family unit information

Because η_{ij} is a single index in f_i with $\nabla_{f_i} \eta_{ij} = \lambda_j$ (Proposition 7), the information that item j carries about the factors factorizes through the scalar index:

$$\mathcal{I}_j(f_i) = \mathcal{J}_j(\eta_{ij}) \lambda_j \lambda_j^\top, \quad \mathcal{J}_j(\eta) = -\mathbb{E}_{X|\eta} \left[\frac{\partial^2}{\partial \eta^2} \log F_j(X | \eta) \right], \quad (\text{S24})$$

so the per-factor information is $\mathcal{I}_{jk}(f_i) = \lambda_{jk}^2 \mathcal{J}_j(\eta_{ij})$ — Eq. (9) of the main text — with $\mathcal{J}_j$ the one-dimensional Fisher information of the item's own likelihood family. We derive $\mathcal{J}_j$ for each family.

Proposition 10 (Family unit informations). *For the five families of (S6)–(S10),*

$$\text{Gaussian: } \mathcal{J}_j(\eta) = \frac{1}{\sigma_j^2}, \quad (\text{S25})$$

$$\text{Bernoulli (logit): } \mathcal{J}_j(\eta) = p(1-p), \quad p = \text{logit}^{-1}(\eta), \quad (\text{S26})$$

$$\text{Negative-binomial (log link): } \mathcal{J}_j(\eta) = \frac{\mu r_j}{\mu + r_j}, \quad \mu = e^\eta, \quad (\text{S27})$$

$$\text{Graded-response ordinal: } \mathcal{J}_j(\eta) = \sum_{c=1}^{C_j} \frac{(\partial_\eta P_{jc}(\eta))^2}{P_{jc}(\eta)} = \sum_{c=1}^{C_j-1} \frac{(P_{jc}^*(1 - P_{jc}^*) - P_{j,c+1}^*(1 - P_{j,c+1}^*))^2}{P_{jc}(\eta)}, \quad (\text{S28})$$

where $P_{jc}(\eta) = P_{jc}^* - P_{j,c+1}^*$ is the probability of category c and $P_{jc}^* = \text{logit}^{-1}(\eta - \tau_{jc})$ is the cumulative probability of exceeding threshold τ_{jc} (with $P_{j0}^* = 1$, $P_{j,C_j}^* = 0$).

Proof. Gaussian. $\log F_j = -\frac{1}{2} \log(2\pi\sigma_j^2) - \frac{(X-\eta)^2}{2\sigma_j^2}$, so $\partial_\eta^2 \log F_j = -1/\sigma_j^2$, a constant; its negative expectation is $1/\sigma_j^2$, independent of η .

Bernoulli. With $p = \text{logit}^{-1}(\eta)$, $\log F_j = X \log p + (1-X) \log(1-p)$ and $\partial_\eta p = p(1-p)$. The score is $\partial_\eta \log F_j = X - p$, and $\mathcal{J}_j = \text{Var}(X - p | \eta) = \text{Var}(X) = p(1-p)$.

Negative-binomial. For $X \sim \text{NegBinom}$ with mean $\mu = e^\eta$ and dispersion r_j , $\text{Var}(X | \eta) = \mu + \mu^2/r_j$. With the log link $\partial_\eta \mu = \mu$, the generalized-linear-model information is $\mathcal{J}_j = (\partial_\eta \mu)^2 / \text{Var}(X) = \mu^2 / (\mu + \mu^2/r_j) = \mu r_j / (\mu + r_j)$.

Graded-response ordinal. For a categorical likelihood $\log F_j = \sum_c \mathbf{1}\{X = c\} \log P_{jc}(\eta)$, the score is $\partial_\eta \log F_j = \sum_c \mathbf{1}\{X = c\} \partial_\eta P_{jc} / P_{jc}$ and the expected squared score is $\mathcal{J}_j = \sum_c (\partial_\eta P_{jc})^2 / P_{jc}$, the standard graded-response information [1]. Writing category probabilities as differences of the cumulative logits P_{jc}^* and using $\partial_\eta P_{jc}^* = P_{jc}^* (1 - P_{jc}^*)$ gives the closed form on the right of (S28). $\square$

S4.3 Why loading is not information

Equation (S24) is why localization and reflection part company. The loading λ_{jk} enters $\mathcal{I}_{jk}$ squared, but it multiplies the unit information $\mathcal{J}_j(\eta_{ij})$, which for every non-Gaussian family depends on where the patient sits. The Bernoulli case (S26) is the sharpest: a binary flag endorsed by a fraction p of patients contributes $\mathcal{J}_j = p(1-p)$, which is 0.09 at $p = 0.1$ and collapses to 0 as $p \rightarrow 0$, *regardless of* λ . A strongly-loading, rarely-endorsed symptom therefore localizes almost no one. Only for the Gaussian family (S25), where $\mathcal{J}_j = 1/\sigma_j^2$ is constant, does the ranking by information coincide with the ranking by (scaled) loading.

Corollary 2 (Reliability and the target-driven item count). *On a single axis k , accumulating (S23) gives a scalar precision $\pi_k = 1 + \sum_j \mathcal{I}_{jk}$ (prior precision 1 for the standardized factor) and a reliability*

$$\rho_k = 1 - \frac{1}{\pi_k} = 1 - \frac{1}{1 + \sum_j \lambda_{jk}^2 \mathcal{J}_j}. \quad (\text{S29})$$

To reach a posterior standard deviation $\pi_k^{-1/2} \leq \varsigma$ requires accumulated information $\sum_j \mathcal{I}_{jk} \geq \varsigma^{-2} - 1$; at $\varsigma = 0.5$ this is $\sum_j \mathcal{I}_{jk} \geq 3$. For a standardized continuous indicator of common loading λ the residual variance is $1 - \lambda^2$, so its Gaussian unit information is $\mathcal{J} = 1/(1 - \lambda^2)$ and it contributes $\lambda^2/(1 - \lambda^2)$ per item; the required count is therefore

$$n \geq \frac{\varsigma^{-2} - 1}{\lambda^2/(1 - \lambda^2)} = 3 \frac{1 - \lambda^2}{\lambda^2} \quad (\varsigma = 0.5), \quad (\text{S30})$$

i.e. 3.1 items at $\lambda = 0.7$, 9.0 at $\lambda = 0.5$ and 30.3 at $\lambda = 0.3$ — consistent with the “roughly 4, 9 and 31 items” the full mixed-bank simulation reports in §3.2.

Proof. Immediate from $\pi_k = 1 / \text{Var}(f_{ik} | X)$ and (S29): the item count solves $\lambda^2 \mathcal{J} \cdot n = \varsigma^{-2} - 1$ with the standardized-indicator unit information $\mathcal{J} = 1/(1 - \lambda^2)$, giving (S30). $\square$

S5 Adaptive sampling and value of information

S5.1 The fixed order as an upper bound on adaptive efficiency

Ordering the candidate items for an axis by their information (S24) and accumulating (S23) gives the reliability-versus-items curve (S29). Evaluated at the population mean $f_i = \mathbf{0}$ this is the efficiency of a fixed, non-adaptive most-informative-first battery. A computerized adaptive test (CAT) instead re-evaluates $\mathcal{J}_j(\eta_{ij})$ at each patient’s provisional estimate $\hat{\theta}$ and selects the currently most informative item.

Proposition 11 (Population-mean order is an upper bound). *Fix an axis and let $\bar{\mathcal{I}}_j = \mathcal{J}_j(\bar{\eta}_j)$ be the unit information evaluated at the population-mean linear predictor $\bar{\eta}_j = \alpha_j$. The fixed schedule that administers items in decreasing $\lambda_j^2 \bar{\mathcal{I}}_j$ maximizes accumulated information at every step among schedules that do not use the patient’s responses. Any adaptive schedule attains, in expectation over patients, no more accumulated information per item at the population mean than this fixed order.*

Proof. At the population mean the per-item increments $\lambda_j^2 \bar{\mathcal{I}}_j$ are fixed constants; by the rearrangement inequality, administering them in decreasing order maximizes the partial sums $\sum_{j \leq n}$ at every n , so the fixed most-informative-first order is optimal among response-independent schedules. An adaptive schedule can exceed a fixed one only by exploiting the dependence of $\mathcal{J}_j(\eta_{ij})$ on the unknown f_i ; at the population mean, and for families whose $\mathcal{J}_j$ does not depend on η (the Gaussian family, (S25)), that dependence is absent and the two schedules coincide. $\square$

This is why the CAT simulation of §3.3 finds the bound *tight*: on every axis whose leading items are Gaussian the adaptive and fixed curves coincide to within 2×10^{-4} in reliability, because $\mathcal{J}_j$ is constant and there is nothing to adapt to; only Bernoulli-topped axes (whose $\mathcal{J}_j = p(1-p)$ varies with θ) show a small adaptive signature, at most 0.012 in reliability. The bank-limited axes are a separate phenomenon: mania/activation and substance plateau at reliability 0.41 and 0.43 because $\sum_j \lambda_{jk}^2 \mathcal{J}_j$ saturates over a 2- and 4-item bank respectively — a ceiling of (S29) set by instrument content, not by any schedule.

S5.2 Value of information: the matrix-determinant lemma

Across axes the same information becomes a design criterion through the entropy of the posterior coordinate.

Proposition 12 (Closed-form entropy reduction). *The differential entropy of the Gaussian posterior is $H(f_i) = \frac{1}{2} \log \det(2\pi e S_i)$. Adding a single not-yet-measured indicator j^* with loading λ_{j^*} and unit information $\mathcal{J}_{j^*} = \mathcal{J}_{j^*}(\hat{\eta}_{ij^*})$ reduces the entropy by*

$$H(f_i) - H(f_i | j^*) = \frac{1}{2} \log \left(1 + \mathcal{J}_{j^*} \lambda_{j^*}^\top S_i \lambda_{j^*} \right), \quad (\text{S31})$$

a closed form that depends on what has already been measured through the current posterior covariance S_i . Equation (S31) is Eq. (10) of the main text.

Proof. By Proposition 9 the updated precision is $S_i^{-1} |_{+j^*} = S_i^{-1} + \mathcal{J}_{j^*} \lambda_{j^*} \lambda_{j^*}^\top$, a rank-one update. The entropy change is $H(f_i) - H(f_i | j^*) = \frac{1}{2} \log \det(S_i^{-1} |_{+j^*}) - \frac{1}{2} \log \det(S_i^{-1}) = \frac{1}{2} \log \frac{\det(S_i^{-1} + \mathcal{J}_{j^*} \lambda_{j^*} \lambda_{j^*}^\top)}{\det S_i^{-1}}$. The matrix-determinant lemma $\det(A + uv^\top) = \det(A)(1 + v^\top A^{-1}u)$ with $A = S_i^{-1}$, $u = \mathcal{J}_{j^*} \lambda_{j^*}$, $v = \lambda_{j^*}$ gives the ratio $1 + \mathcal{J}_{j^*} \lambda_{j^*}^\top S_i \lambda_{j^*}$, and the logarithm halves to (S31). $\square$

Two properties make (S31) the right design criterion. It is *context-dependent*: the value of a candidate item scales with $\lambda_{j^*}^\top S_i \lambda_{j^*}$, the current posterior variance along the item’s loading direction, so an item is worth most on an axis that is still uncertain and its value falls as that axis fills in. And it is *diminishing*: because S_i shrinks with each added item, the marginal entropy reduction of the next item decreases, so a greedy rule spreads measurement across axes rather than piling onto one. Cycling the greedy selection across the eight axes assembles the shared 27-item battery at mean reliability 0.70 of §3.4; applied within a cohort, the same S_i -weighted criterion localizes where an axis is under-measured (e.g. the schizophrenia immunometabolic and sleep deficits, 20.7 vs 34.9 and 3.9 vs 8.4 items).

Corollary 3 (Greedy value is a lower bound on the jointly-optimal design). *The entropy reduction $f \mapsto \log \det(\Phi^{-1} + \sum_{j \in S} \mathcal{I}_j)$ as a function of the selected set S is monotone and submodular, so the greedy battery attains at least a $(1 - 1/e)$ fraction of the entropy reduction of the jointly-optimal same-size battery.*

Proof. Monotonicity holds because each $\mathcal{I}_j \succeq 0$ (Proposition 9) can only increase the log-determinant. Submodularity of $S \mapsto \log \det(A + \sum_{j \in S} \mathcal{I}_j)$ for positive semidefinite $\mathcal{I}_j$ is the standard concavity-of-log-det result: the marginal gain of adding j^* , given by (S31), is non-increasing as the base set grows because S_i (the inverse of the accumulated precision) is decreasing

in the Loewner order. The $(1 - 1/e)$ guarantee is the Nemhauser–Wolsey–Fisher bound for greedy maximization of a monotone submodular function. This bounds the price of greediness noted as a limitation in the main text. $\square$

S6 Archetypes as a low-dimensional summary and its reconstruction fidelity

S6.1 The archetypal-analysis program

The cloud of posterior coordinates $\{x_i\}_{i=1}^N$, $x_i = \hat{f}_i \in \mathbb{R}^{K+1}$, is summarized by archetypal analysis [2]: each patient is written as a convex blend of A extreme phenotypes that are themselves convex combinations of patients,

$$\min_{W,B} \sum_{i=1}^N \left\| x_i - \sum_{a=1}^A w_{ia} z_a \right\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^N b_{ja} x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \sum_a w_{ia} = \sum_j b_{ja} = 1, \quad (\text{S32})$$

i.e. $X \approx WZ$ with archetypes $Z = BX$ and both W (patient $\times$ archetype) and B (patient $\times$ archetype) row-stochastic. This is Eq. (11) of the main text. The constraints place each archetype z_a inside the convex hull of the data and each patient’s reconstruction $\hat{x}_i = \sum_a w_{ia} z_a$ inside the convex hull of the archetypes; the simplex weight $w_i \in \Delta^{A-1}$ is the patient’s soft membership. The number of corners is fixed by stability: $A = 5$ is the largest with cross-seed Tucker congruence ≥ 0.8 , with a cliff at $A = 6$, and w_i is propagated across posterior draws.

S6.2 Reconstruction fidelity as a variance decomposition

Because (S32) is a rank- A convex approximation it discards information, and unlike an unconstrained projection its residual is not orthogonal to the fitted part. We quantify the loss by a coefficient of determination.

Definition 1 (Reconstruction R^2). With reconstruction $\hat{x}_i = \sum_a w_{ia} z_a$, the per-axis and pooled reconstruction R^2 are

$$R_k^2 = 1 - \frac{\sum_i (x_{ik} - \hat{x}_{ik})^2}{\sum_i (x_{ik} - \bar{x}_k)^2}, \quad R_{\text{pool}}^2 = 1 - \frac{\sum_i \|x_i - \hat{x}_i\|^2}{\sum_i \|x_i - \bar{x}\|^2} = \frac{\sum_k \text{SS}_k^{\text{tot}} R_k^2}{\sum_k \text{SS}_k^{\text{tot}}}, \quad (\text{S33})$$

the pooled value being the variance-weighted mean of the per-axis values with weights $\text{SS}_k^{\text{tot}} = \sum_i (x_{ik} - \bar{x}_k)^2$.

Evaluated on all $N = 9,013$ patients the five-corner blend attains $R_{\text{pool}}^2 = 0.590$: the convex summary retains 59% of the coordinate variance. It is tempting to read this against the unconstrained linear optimum at the same dimension — a five-component principal-component projection attains $R^2 = 0.796$, a dimension-matched four-component projection 0.680 — and conclude that convex interpretability costs 20.6 percentage points of fidelity. That reading is wrong, and Proposition 13 below explains why: on this map the coordinate variance is nearly isotropic, so reconstruction R^2 is an uninformative test of any five-dimensional summary. A *random* five-dimensional subspace retains an expected $k/d = 5/8 = 0.625$ of the variance regardless of the covariance spectrum — more than the archetype simplex — and a 500-seed simulation confirms it (0.624 ± 0.042 , placing the archetype value at the 21st percentile of random projections). The simplex does not merely trail PCA; it trails a coin-flip subspace, because variance capture is not what it optimizes. What it does provide, and what neither PCA nor a random subspace can, is convex non-negative sum-to-one membership in A *named* corners: a principal component is a signed axis-mixing contrast that no patient can equal, whereas each archetype z_a is a near-real extreme profile a patient can be a stated fraction of. Reconstruction

R^2 is therefore reported here as an honesty check on how much of the continuum the corners span — not as a competitive metric — and the five-centroid k -means hard partition ($R^2 = 0.325$), the discrete alternative appropriate only if the cloud had clusters, is included to show that the least defensible summary is the categorical one. Per axis, R_k^2 is high where the corners separate (immunometabolic 0.89, overall severity 0.86, sleep 0.76, developmental risk 0.75) and near-flat on the two bank-limited axes (cognition 0.06, substance ≈ 0), the same axes flagged by (S29) — so the per-axis profile is a diagnostic of where the instrument is weak, not a scorecard.

Proposition 13 (A random k -subspace captures k/d of the variance). *Let $x_1, \dots, x_N \in \mathbb{R}^d$ have mean zero and total variance $T = \frac{1}{N} \sum_i \|x_i\|^2$. Let P be the orthogonal projector onto a k -dimensional subspace drawn uniformly (Haar) from the Grassmannian $\text{Gr}(k, d)$. Then the expected variance captured by projecting the data onto that subspace is*

$$\mathbb{E}\left[\frac{1}{N} \sum_i \|Px_i\|^2\right] = \frac{k}{d} T, \quad \text{i.e.} \quad \mathbb{E}[R_{\text{rand-}k}^2] = \frac{k}{d}, \quad (\text{S34})$$

independent of the eigenvalue spectrum of the data covariance.

Proof. Write the covariance $\Sigma = \frac{1}{N} \sum_i x_i x_i^\top$ with eigenvalues $\sigma_1^2 \geq \dots \geq \sigma_d^2$ summing to T . The captured variance is $\frac{1}{N} \sum_i x_i^\top P x_i = \text{tr}(P\Sigma)$. For a Haar-random rank- k projector, $\mathbb{E}[P] = (k/d)I_d$ by rotational invariance (the only symmetric matrix commuting with all rotations is a scalar multiple of the identity, and $\text{tr} \mathbb{E}[P] = k$ fixes the scalar). Hence $\mathbb{E}[\text{tr}(P\Sigma)] = \text{tr}((k/d)\Sigma) = (k/d)T$, and dividing by T gives (S34). The result uses only $\mathbb{E}[P] = (k/d)I$; the spectrum $\{\sigma_j^2\}$ cancels, which is why an anisotropic (e.g. single-dominant-component) cloud and a near-isotropic one both give k/d in expectation, while their PCA-optimal R^2 differ sharply. $\square$

For this map $k/d = 5/8 = 0.625$; the simulated mean 0.624 matches to Monte-Carlo error, and the gap between that and the PCA-5 value 0.796 is a direct measure of how far the coordinate cloud departs from isotropy — here, modestly (the leading component carries 0.269 of the variance, against 0.125 for an exactly isotropic cloud).

Proposition 14 (Fidelity does not degrade for ambiguous patients). *Let $e_i = \|x_i - \hat{x}_i\|$ be a patient’s reconstruction error and $H_i = -\sum_a w_{ia} \log w_{ia}$ the entropy of their archetype weights. Then $\text{Corr}(e_i, H_i) < 0$: interior “fog” patients reconstruct better, not worse.*

Argument. By construction each $\hat{x}_i$ lies in the convex hull of the archetypes. A high-entropy patient has weights spread across corners and therefore sits near the centroid of the hull, where the convex approximation is tight; a low-entropy (near-corner) patient sits near a single archetype z_a , but the true x_i near a corner frequently lies *outside* the hull spanned by the finite archetype set (the archetypes are interior convex combinations of data, not the extreme points themselves), so the clipped reconstruction incurs a larger residual. Empirically $\text{Corr}(e_i, H_i) = -0.13$ ($p < 10^{-30}$), consistent with this geometry. The negative sign is the useful conclusion: the summary is not purchased by sacrificing the ambiguous patients it is meant to describe. $\square$

S7 Calibration protocol

The instrument reports posterior intervals; §2.7 tests whether those intervals are honest. Because the truth is unknown for real patients, coverage is tested by simulation under the model’s own generative law with real missingness.

S7.1 Synthetic-truth generative design

We draw $M = 10,000$ synthetic patients (seed 20260704) with known latent coordinate $\theta_m \sim \mathcal{N}(\mathbf{0}, \Phi)$ under the frozen model’s factor prior. For each we generate item responses from

the fitted families (S6)–(S10) at $\eta_{mj} = \alpha_j + \lambda_j^\top \theta_m$, and impose a real missingness pattern by bootstrapping the observed/missing masks of the 9,013 real patients — only the binary observed-or-missing indicator is used, never any clinical value. This preserves the empirical joint distribution of *which* axes are co-measured, which is what stresses the projection.

S7.2 Fisher-scoring EAP and its exactness on Gaussian patterns

Each synthetic patient is projected by the same machinery as a real one. For non-Gaussian patterns the posterior mode and curvature are found by Fisher scoring: starting from $\hat{\theta}^{(0)} = \mathbf{0}$, iterate

$$\hat{\theta}^{(t+1)} = \hat{\theta}^{(t)} + \left(\Phi^{-1} + \sum_{j \in O_m} \mathcal{I}_j(\hat{\theta}^{(t)}) \right)^{-1} \left(-\Phi^{-1} \hat{\theta}^{(t)} + \sum_{j \in O_m} u_{mj}(\hat{\theta}^{(t)}) \right), \quad (\text{S35})$$

where $u_{mj} = \lambda_j \partial_\eta \log F_j(X_{mj} | \eta_{mj})$ is the item score and $\mathcal{I}_j$ its expected information (S24); the posterior covariance is the inverse information $S_m = (\Phi^{-1} + \sum_j \mathcal{I}_j(\hat{\theta}))^{-1}$ at convergence, the Laplace approximation to the posterior.

Proposition 15 (Exactness on Gaussian patterns). *If all observed items of patient m are Gaussian (S6), the Fisher-scoring iteration (S35) converges in one step from any start to the closed-form EAP (S19), and S_m equals the closed-form posterior covariance exactly.*

Proof. For a Gaussian item $\mathcal{I}_j = \lambda_j \lambda_j^\top / \sigma_j^2$ is independent of θ and the score is affine, $u_{mj}(\theta) = \lambda_j(X_{mj} - \alpha_j - \lambda_j^\top \theta) / \sigma_j^2$. The bracketed Newton system (S35) is then exactly the normal equations of the Gaussian posterior, whose solution is (S20)–(S19); a Newton step on a quadratic objective reaches the optimum in one iteration regardless of the starting point. $\square$

We verified this numerically: on all-Gaussian patterns the Fisher-scoring coordinate matches the closed form (S19) to 3×10^{-15} , so the calibration study exercises the *same* estimator the instrument uses, extended to the non-Gaussian families.

S7.3 Coverage statistics and the prior-in-prior-out check

For nominal levels $q \in \{0.50, 0.80, 0.90, 0.95\}$ with standard-normal quantile z_q , we report three statistics: the per-axis marginal coverage

$$\widehat{\text{cov}}_k(q) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\left\{ |\theta_{mk} - \hat{x}_{mk}| \leq z_{(1+q)/2} [S_m]_{kk}^{1/2} \right\}, \quad (\text{S36})$$

the joint coverage of the 8-dimensional credible ellipsoid via the Mahalanobis distance against a χ_{K+1}^2 quantile, $\widehat{\text{cov}}_{\text{joint}}(q) = \frac{1}{M} \sum_m \mathbf{1}\{(\theta_m - \hat{x}_m)^\top S_m^{-1}(\theta_m - \hat{x}_m) \leq \chi_{K+1,q}^2\}$, and the 95% coverage stratified by the number of observed indicators on the axis. The Monte-Carlo standard error of a coverage estimate at level q is $\sqrt{q(1-q)/M}$; at $q = 0.95$, $M = 10^4$ this is 0.0022.

Proposition 16 (Prior in, prior out). *On an axis k with no observed indicator, the projection returns the prior on that coordinate — $\hat{x}_{mk} = 0$ and $[S_m]_{kk}$ equal to the prior variance — so the coverage (S36) equals the nominal level exactly, up to Monte-Carlo error.*

Proof. With no observed item loading on axis k , the corresponding rows of Λ_{O_m} are zero, so the data term $\sum_j \mathcal{I}_j$ in (S23) has no entry on coordinate k and the posterior marginal of θ_{mk} equals its prior $\mathcal{N}(0, \Phi_{kk})$. Since the synthetic truth is drawn from that same prior (S1), the interval $0 \pm z_{(1+q)/2} \Phi_{kk}^{1/2}$ covers θ_{mk} with probability exactly q . $\square$

This makes the zero-item stratum a built-in correctness check rather than a free parameter: the instrument *must* recover nominal coverage there, and it does. Across all strata the empirical coverage tracks nominal — 0.501, 0.801, 0.900, 0.949 against 0.50/0.80/0.90/0.95, joint ellipsoid 0.947 at 95% — and is flat from the prior-dominated stratum to the richly-observed one (0.947–0.952 across item-count bins), the result reported in §3.5.

What the study does and does not establish. Coverage here demonstrates that the projection machinery is *internally honest*: its 95% intervals cover the truth 95% of the time when the data are generated by the fitted model under real missingness. It does not establish that the fitted model is the true data-generating process — no coverage study can, since it conditions on the generative law. On the discrete-informed axes the Laplace posterior standard deviation runs slightly below the residual error (a light-tailed approximation), so interval *width* is marginally optimistic even though the 95%-level coverage, fixed by the interval quantiles, is exact. The next subsection addresses the conditioning limitation directly.

S7.4 Robustness of coverage under misspecification

The coverage study of §S7.3 draws data from the fitted model, so it cannot distinguish a well-calibrated estimator from one that is calibrated only because it scores data it also generated. We therefore stress the projection with data drawn from processes it does *not* assume, and ask how far coverage drifts. Four one-parameter families of violations are used, each targeting a specific assumption of §S1:

1. **Correlated residuals** (violates local independence, (S4)): the residual of every item receives a shared nuisance component, $\varepsilon_j \leftarrow (\varepsilon_j + \rho g_*) / \sqrt{1 + \rho^2}$ with common $g_* \sim \mathcal{N}(0, 1)$ and $\rho \in \{0, 0.15, 0.30, 0.45\}$.
2. **Omitted factor** (violates the rank- K assumption): a ninth latent $g_9 \sim \mathcal{N}(0, 1)$ loads a random 40% of items with strength $s \in \{0, 0.25, 0.5, 0.75\}$, absent from the scoring model.
3. **Loading mismatch** (parameter error / transfer to a new sample): the scoring model uses $\lambda_{jk}(1 + \varepsilon\xi_{jk})$, $\xi_{jk} \sim \mathcal{N}(0, 1)$, $\varepsilon \in \{0, 0.10, 0.20, 0.30\}$, while data use the true λ_{jk} .
4. **Heavy tails** (violates Gaussian residuals): continuous residuals are Student- t with $\text{df} \in \{\infty, 8, 5, 3\}$, rescaled to unit variance.

Each cell draws $N = 4,000$ synthetic patients keeping 50% of items (three seeds), scores them with the *unperturbed* Gaussian EAP, and records empirical coverage of the nominal 95% per-axis interval. The results (Fig. 9B; §3.8) show graceful, monotone degradation and no collapse:

Violation	none	low	med	high
Correlated residuals (ρ)	0.951	0.943	0.918	0.889
Omitted ninth factor (s)	0.951	0.948	0.939	0.939
Loading mismatch (ε)	0.949	0.950	0.927	0.934
Heavy-tailed residuals (df)	0.949	0.950	0.950	0.955

Three features are worth noting. First, the unperturbed baseline recovers nominal (0.949–0.951) in every family, confirming the estimator is unbiased when its assumptions hold. Second, heavy tails are absorbed essentially without loss — indeed slightly conservatively at the 50% level — because the EAP posterior variance depends on the loadings and the residual *variance*, not on higher residual moments. Third, the steepest degradation is under correlated residuals, and this is expected: local dependence inflates the effective information a naive independent-likelihood score believes it has, shrinking intervals below their honest width. Even so, an extreme shared-nuisance loading ($\rho = 0.45$, a residual correlation the model-selection diagnostics of the main text would flag) costs only six points of coverage. The projection’s intervals are thus honest under the model and degrade predictably, not catastrophically, away from it; local dependence is the assumption whose violation matters most, and the one most worth checking on real data before trusting interval *widths*.

Summary of cross-references to the main text

Each numbered result here underlies a specific claim in the main manuscript: (S11) and Proposition 1 justify the no-imputation invariant (Eq. (3)); Lemma 1 and Proposition 4 are the projection (Eq. (6), Eq. (7)); Propositions 7–8 are the loading-as-slope (Eq. (5)); Propositions 9–10 and Corollary 2 are the information accounting (Eq. (8), Eq. (9)); Propositions 11–12 and Corollary 3 are adaptive sampling and value of information (Eq. (10)); Definition 1, Proposition 13 (the random-subspace k/d bound that reframes reconstruction R^2) and Proposition 14 are the archetype reconstruction; and Propositions 15–16 with the misspecification stress test of §S7.4 are the calibration protocol and its robustness (§2.7, §3.8).

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